# LAB NOC 03

## Cisco Catalyst 2960 — EtherChannel Failure / VLAN Mismatch Troubleshooting

Real Hardware | ES / EN | NOC Portfolio

### INFORMACION DEL LABORATORIO / LAB INFORMATION

| | | |
|---|---|---|
| **Hardware / Hardware** | Cisco Catalyst 2960 (WS-C2960-24TT-L) | Cisco Catalyst 2960 (WS-C2960-24TT-L) |
| **Categoria / Category** | Troubleshooting EtherChannel / VLAN | EtherChannel / VLAN Troubleshooting |
| **Nivel / Level** | CCNA — Switching / Troubleshooting | CCNA — Switching / Troubleshooting |
| **Duracion / Duration** | Aprox. 60 minutos | Approx. 60 minutes |
| **Entorno / Environment** | Hardware real (equipo usado) | Real hardware (used equipment) |
| **Tecnologias** | VLAN, EtherChannel, Forced mode, Troubleshooting | VLAN, EtherChannel, Forced mode, Troubleshooting |
| **Relevancia NOC** | Resolucion de incidente L2, cierre de ticket | L2 incident resolution, ticket closure |

### OBJETIVO / OBJECTIVE

| OBJETIVO | OBJECTIVE |
|---|---|
| Diagnosticar y resolver una falla de EtherChannel causada por VLAN mismatch en un Cisco Catalyst 2960 real. Documentar el incidente en formato NOC (Incident Report), ejecutar los pasos de recuperacion y verificar el restablecimiento del servicio. | Diagnose and resolve an EtherChannel failure caused by VLAN mismatch on a real Cisco Catalyst 2960. Document the incident in NOC format (Incident Report), execute recovery steps, and verify service restoration. |

### PROCEDIMIENTO — 4 ETAPAS / PROCEDURE — 4 STAGES

| # | ESPANOL | ENGLISH |
|---|---|---|
| 1 | ETAPA 1 — Baseline: verificar estado inicial del switch | STAGE 1 — Baseline: verify initial switch state |
| 2 | Ejecutar: show vlan brief → confirmar VLAN 1 activa, VLANs 1002-1005 reservadas | Run: show vlan brief → confirm VLAN 1 active, VLANs 1002-1005 reserved |
| 3 | Ejecutar: show etherchannel summary → detectar Port-Channel1 (Po1) preexistente del lab anterior | Run: show etherchannel summary → detect pre-existing Port-Channel1 (Po1) from previous lab |
| 4 | Ejecutar: show interfaces status → Fa0/6 suspended VLAN 10, Fa0/16 suspended VLAN | Run: show interfaces status → Fa0/6 suspended VLAN 10, Fa0/16 suspended VLAN |

| | | |
|---|---|---|
| | 20, Po1 DOWN | 20, Po1 DOWN |
| 5 | ETAPA 2 — Diagnostico: identificar causa raiz | STAGE 2 — Diagnosis: identify root cause |
| 6 | Identificar error en syslog: EC-5-CANNOT_BUNDLE2 — Fa0/6 not compatible with Po1 | Identify syslog error: EC-5-CANNOT_BUNDLE2 — Fa0/6 not compatible with Po1 |
| 7 | Confirmar: Fa0/6 en VLAN 10 y Fa0/16 en VLAN 20 — VLAN mismatch entre miembros del bundle | Confirm: Fa0/6 on VLAN 10 and Fa0/16 on VLAN 20 — VLAN mismatch between bundle members |
| 8 | Causa raiz verificada: EtherChannel requiere configuracion identica en todos sus puertos miembro | Root cause verified: EtherChannel requires identical configuration on all member ports |
| 9 | ETAPA 3 — Resolucion: recuperar el servicio | STAGE 3 — Resolution: restore service |
| 10 | default interface fa0/6 y fa0/16 — resetear puertos a configuracion de fabrica | default interface fa0/6 and fa0/16 — reset ports to factory configuration |
| 11 | no interface port-channel 1 — eliminar Po1 (puertos pasan a administratively down) | no interface port-channel 1 — delete Po1 (ports go administratively down) |
| 12 | Reconfigurar Fa0/6 y Fa0/16: mode access, VLAN 10 en ambos, channel-group 1 mode on | Reconfigure Fa0/6 and Fa0/16: mode access, VLAN 10 on both, channel-group 1 mode on |
| 13 | ETAPA 4 — Verificacion: show etherchannel summary → Po1 estado U (in use), puertos activos | STAGE 4 — Verification: show etherchannel summary → Po1 state U (in use), ports active |
| 14 | Confirmar en syslog: LINEPROTO-5-UPDOWN: Interface Fa0/6 changed state to up (idem Fa0/16) | Confirm in syslog: LINEPROTO-5-UPDOWN: Interface Fa0/6 changed state to up (same Fa0/16) |

## Baseline del Switch antes del LAB NOC 03

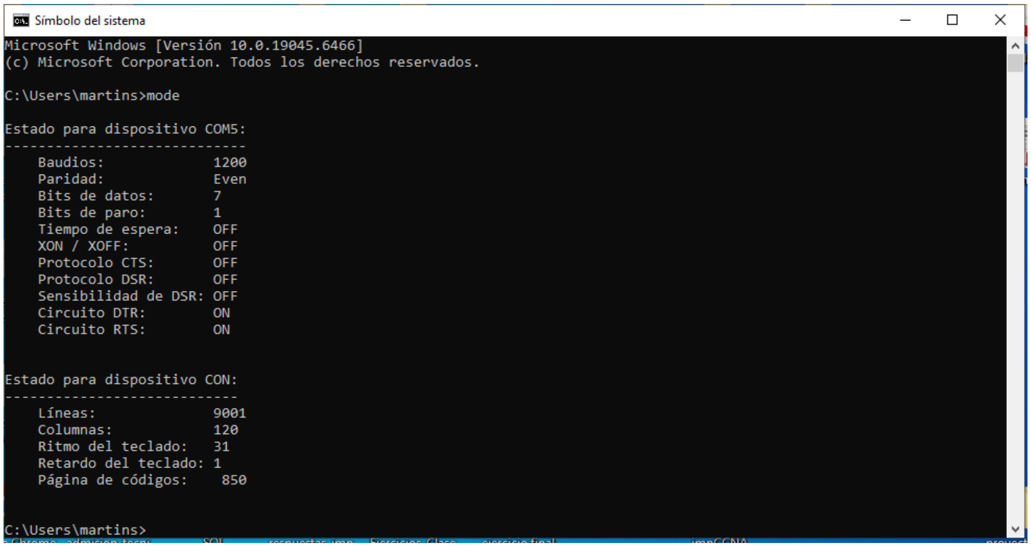
**Equipo:** Cisco Catalyst 2960
**Estado inicial observado:**

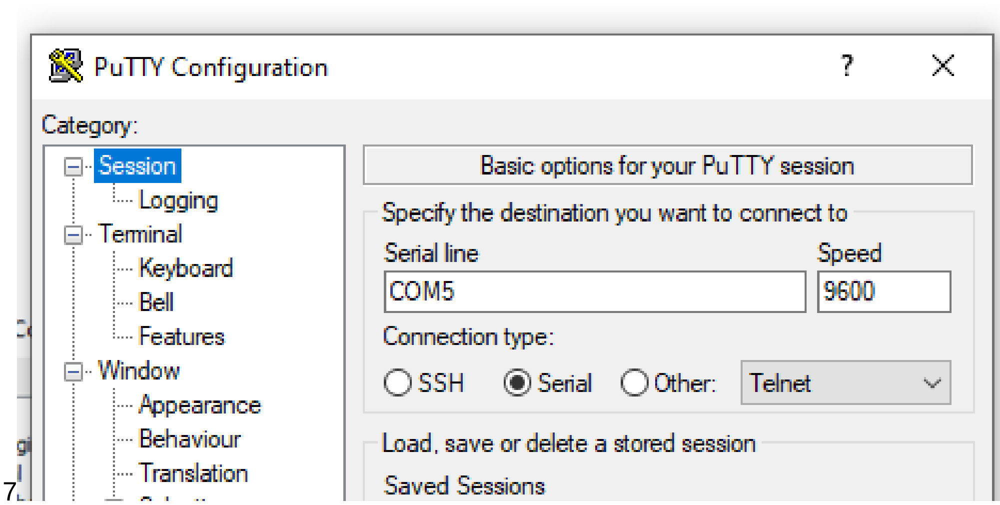
- VLAN 1 activa
- VLAN 1002–1005 reservadas
- **Port-Channel1 existente (Po1)** proveniente del **LAB NOC 02 – EtherChannel**
- Puertos Fa0/6 y Fa0/16 conectados físicamente
- Acceso por **console COM5 vía PuTTY**

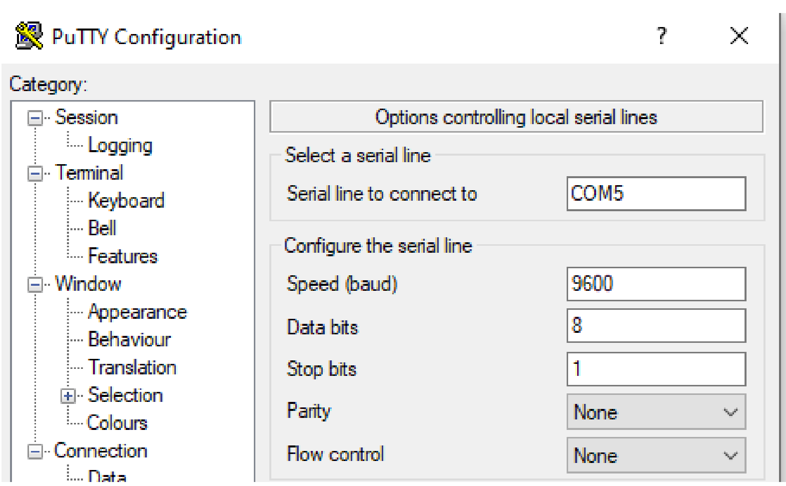
**Baseline verification:**

show vlan brief
show etherchannel summary
show interface status

Observación:
Se detecta Port-Channel1 (Po1) configurado previamente
correspondiente al laboratorio EtherChannel.


Símbolo del sistema
Microsoft Windows [Versión 10.0.19045.6466]
(c) Microsoft Corporation. Todos los derechos reservados.
C:\Users\martins>mode
Estado para dispositivo COM5:
---------------------------
Baudios: 1200
Paridad: Even
Bits de datos: 7
Bits de paro: 1
Tiempo de espera: OFF
XON / XOFF: OFF
Protocolo CTS: OFF
Protocolo DSR: OFF
Sensibilidad de DSR: OFF
Circuito DTR: ON
Circuito RTS: ON
Estado para dispositivo CON:
--------------------------
Líneas: 9001
Columnas: 120
Ritmo del teclado: 31
Retardo del teclado: 1
Página de códigos: 850
C:\Users\martins>


PuTTY Configuration
Category:
Session
Logging
Terminal
Keyboard
Bell
Features
Window
Appearance
Behaviour
Translation
Basic options for your PuTTY session
Specify the destination you want to connect to
Serial line
Speed
COM5
9600
Connection type:
SSH
Serial
Other:
Telnet
Load, save or delete a stored session
Saved Sessions

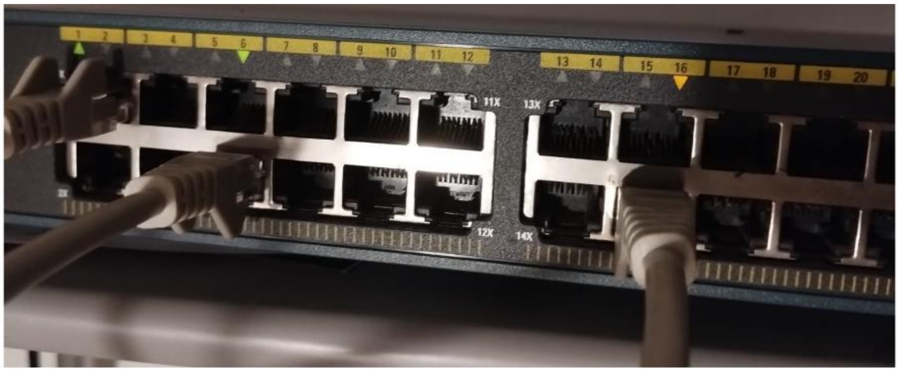

PuTTY Configuration
Category:
Session
Logging
Terminal
Keyboard
Bell
Features
Window
Appearance
Behaviour
Translation
Selection
Colours
Connection
Data
Options controlling local serial lines
Select a serial line
Serial line to connect to
COM5
Configure the serial line
Speed (baud)
9600
Data bits
8
Stop bits
1
Parity
None
Flow control
None

COM5 - PuTTY

```
pF
Switch>
Switch>
Switch>en
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gi0/1, Gi0/2
                                                Po1
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
Switch#
```

```
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name ADMIN
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 20
Switch(config-vlan)#name USERS
Switch(config-vlan)#exit
```

```
Switch(config)#do show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gi0/1, Gi0/2
                                                Po1
10   ADMIN                            active
20   USERS                            active
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
Switch(config)#
```



Physical connection of switch interfaces (Fa0/6 and Fa0/16) before configuration changes

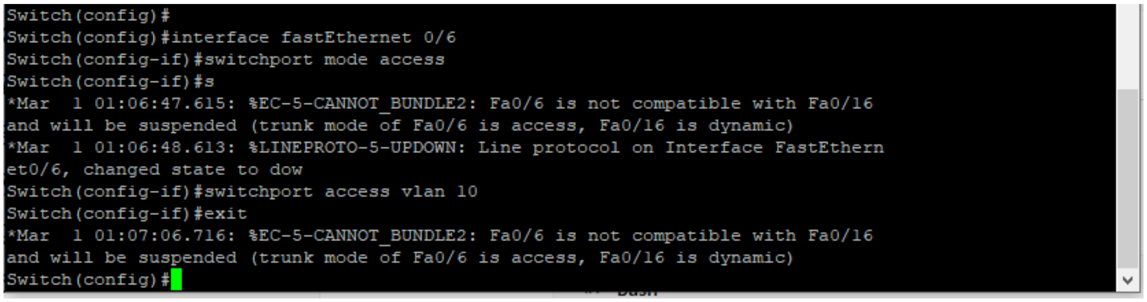
## INCIDENT REPORT — FORMATO NOC / NOC FORMAT

| CAMPO / FIELD | DETALLE / DETAIL |
|---|---|
| **Device / Equipo** | Cisco Catalyst 2960 (WS-C2960-24TT-L) |
| **Ports Affected / Puertos afectados** | Fa0/6, Fa0/16 |
| **Service Affected / Servicio afectado** | Layer 2 connectivity — EtherChannel bundle Po1 |
| **Symptoms / Sintomas** | Puertos muestran estado 'suspended' / Ports show 'suspended' status |
| **Root Cause / Causa raiz** | VLAN mismatch entre puertos miembros del EtherChannel / VLAN mismatch between EtherChannel member ports |
| **Error Message** | EC-5-CANNOT_BUNDLE2: Fa0/6 is not compatible with Po1 |
| **Resolution** | default interface fa0/6 + fa0/16 → no interface port-channel 1 → reconfigure with matching VLAN → channel-group 1 mode on |

## REFERENCIA DE COMANDOS / COMMAND REFERENCE

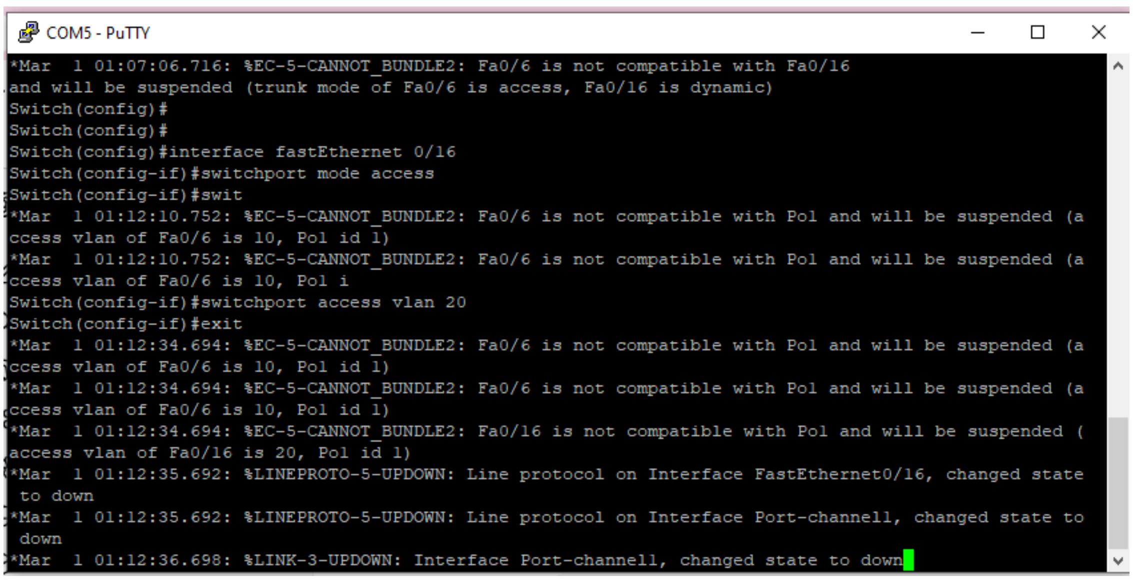
| COMANDO / COMMAND | SIGNIFICADO (ES) | MEANING (EN) |
|---|---|---|
| `show vlan brief` | Verifica VLANs activas y asignacion de puertos | Verifies active VLANs and port assignments |
| `show interfaces status` | Muestra estado de cada puerto (connected/suspended/disabled) | Shows each port status (connected/suspended/disabled) |
| `show etherchannel summary` | Muestra estado del bundle EtherChannel y sus miembros | Shows EtherChannel bundle status and member ports |
| `default interface fa0/X` | Resetea el puerto a configuracion de fabrica | Resets the port to factory configuration |
| `no interface port-channel 1` | Elimina el Port-Channel y libera los puertos miembro | Deletes the Port-Channel and releases member ports |
| `channel-group 1 mode on` | Fuerza la formacion del EtherChannel (sin LACP/PAgP) | Forces EtherChannel formation (without LACP/PAgP) |
| `switchport mode access` | Configura el puerto en modo | Sets port to access mode |

| | | |
|---|---|---|
| | acceso | |
| **`switchport access vlan 10`** | Asigna el puerto a VLAN 10 | Assigns port to VLAN 10 |



```
Switch(config)#
Switch(config)#interface fastEthernet 0/6
Switch(config-if)#switchport mode access
Switch(config-if)#s
*Mar  1 01:06:47.615: %EC-5-CANNOT_BUNDLE2: Fa0/6 is not compatible with Fa0/16
and will be suspended (trunk mode of Fa0/6 is access, Fa0/16 is dynamic)
*Mar  1 01:06:48.613: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/6, changed state to dow
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
*Mar  1 01:07:06.716: %EC-5-CANNOT_BUNDLE2: Fa0/6 is not compatible with Fa0/16
and will be suspended (trunk mode of Fa0/6 is access, Fa0/16 is dynamic)
Switch(config)#
```

Interfaces are incompatible: Fa0/6 is in access mode and Fa0/16 is in dynamic mode.
Fa0/6 = access
Fa0/16 = dynamic



```
COM5 - PuTTY
*Mar  1 01:07:06.716: %EC-5-CANNOT_BUNDLE2: Fa0/6 is not compatible with Fa0/16
and will be suspended (trunk mode of Fa0/6 is access, Fa0/16 is dynamic)
Switch(config)#
Switch(config)#
Switch(config)#interface fastEthernet 0/16
Switch(config-if)#switchport mode access
Switch(config-if)#swit
*Mar  1 01:12:10.752: %EC-5-CANNOT_BUNDLE2: Fa0/6 is not compatible with Po1 and will be suspended (a
ccess vlan of Fa0/6 is 10, Po1 id 1)
*Mar  1 01:12:10.752: %EC-5-CANNOT_BUNDLE2: Fa0/6 is not compatible with Po1 and will be suspended (a
ccess vlan of Fa0/6 is 10, Po1 i
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
*Mar  1 01:12:34.694: %EC-5-CANNOT_BUNDLE2: Fa0/6 is not compatible with Po1 and will be suspended (a
ccess vlan of Fa0/6 is 10, Po1 id 1)
*Mar  1 01:12:34.694: %EC-5-CANNOT_BUNDLE2: Fa0/6 is not compatible with Po1 and will be suspended (a
ccess vlan of Fa0/6 is 10, Po1 id 1)
*Mar  1 01:12:34.694: %EC-5-CANNOT_BUNDLE2: Fa0/16 is not compatible with Po1 and will be suspended (
access vlan of Fa0/16 is 20, Po1 id 1)
*Mar  1 01:12:35.692: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/16, changed state
 to down
*Mar  1 01:12:35.692: %LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to
 down
*Mar  1 01:12:36.698: %LINK-3-UPDOWN: Interface Port-channel1, changed state to down
```

Fa0/6 = access
Fa0/16 = access
A second issue appeared where both Fa0/6 and Fa0/16 were incompatible with Port-Channel 1 (Po1).

This indicates that the existing Port-Channel had a previous configuration that did not match the current settings applied to the interfaces.

Even after correcting the port mode mismatch (changing Fa0/16 from dynamic to access), the EtherChannel could not be formed because Po1 still contained outdated parameters.

To fully resolve the issue, the Port-Channel needed to be removed and the interfaces reset before reconfiguring them with identical settings.

COM5 - PuTTY

```
access vlan of Fa0/16 is 20, Po1 id 1)
*Mar  1 01:12:35.692: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/16, changed state
 to down
*Mar  1 01:12:35.692: %LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to
 down
*Mar  1 01:12:36.698: %LINK-3-UPDOWN: Interface Port-channel1, changed state to down
Switch(config)#
Switch(config)#do show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gi0/1, Gi0/2
10   ADMIN                            active    Fa0/6
20   USERS                            active    Fa0/16
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
Switch(config)#
```

COM5 - PuTTY

```
Switch(config)#do show interfaces status

Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    1          a-full  a-100 10/100BaseTX
Fa0/2                        notconnect   1            auto   auto 10/100BaseTX
Fa0/3                        notconnect   1            auto   auto 10/100BaseTX
Fa0/4                        notconnect   1            auto   auto 10/100BaseTX
Fa0/5                        notconnect   1            auto   auto 10/100BaseTX
Fa0/6                        suspended    10         a-full  a-100 10/100BaseTX
Fa0/7                        notconnect   1            auto   auto 10/100BaseTX
Fa0/8                        notconnect   1            auto   auto 10/100BaseTX
Fa0/9                        notconnect   1            auto   auto 10/100BaseTX
Fa0/10                       notconnect   1            auto   auto 10/100BaseTX
Fa0/11                       notconnect   1            auto   auto 10/100BaseTX
Fa0/12                       notconnect   1            auto   auto 10/100BaseTX
Fa0/13                       notconnect   1            auto   auto 10/100BaseTX
Fa0/14                       notconnect   1            auto   auto 10/100BaseTX
Fa0/15                       notconnect   1            auto   auto 10/100BaseTX
Fa0/16                       suspended    20         a-full  a-100 10/100BaseTX
Fa0/17                       notconnect   1            auto   auto 10/100BaseTX
Fa0/18                       notconnect   1            auto   auto 10/100BaseTX
Fa0/19                       notconnect   1            auto   auto 10/100BaseTX
Fa0/20                       notconnect   1            auto   auto 10/100BaseTX
Fa0/21                       notconnect   1            auto   auto 10/100BaseTX
```

COM5 - PuTTY

```
Gi0/2                        notconnect   1            auto   auto 10/100/1000BaseTX
Po1                          notconnect   1            auto   auto
Switch(config)#
Switch(config)#do show etherchannel summary
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
1      Po1(SD)           -        Fa0/6(s)    Fa0/16(s)

Switch(config)#
```

- **Fa0/6 → suspended → VLAN 10**
- **Fa0/16 → suspended → VLAN 20**
- **Port-channel 1 está DOWN**
  los puertos están **suspendidos**
- Pero en **EtherChannel todos los puertos deben tener la misma configuración**.
- Entonces el switch protege la red y:
- **desactiva el bundle**.
- EC-5-CANNOT_BUNDLE2

**INCIDENT REPORT**

Device: Cisco Catalyst 2960
Ports affected: Fa0/6, Fa0/16
Service affected: Layer2 connectivity

Symptoms:
**Ports show "suspended" status**

Commands used:
**show interfaces status**
**show etherchannel summary**
**show vlan brief**

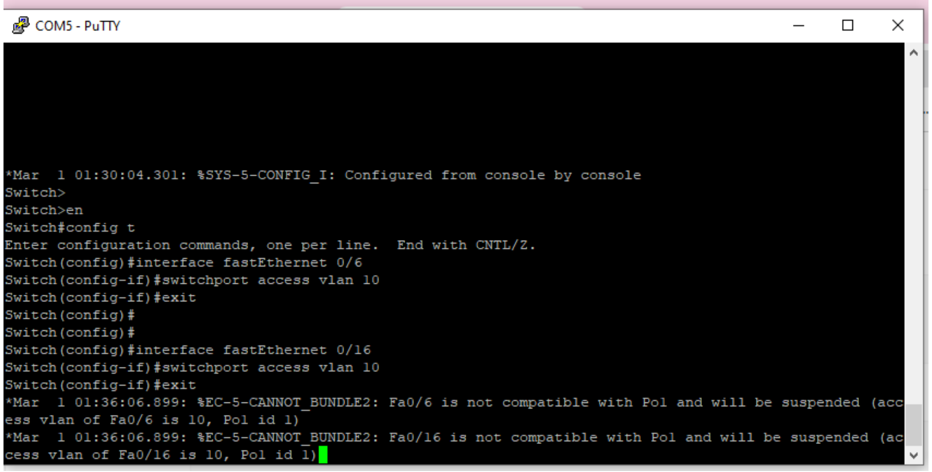
Root cause:
Access VLAN mismatch between EtherChannel member ports

Result:
Port-channel1 went down

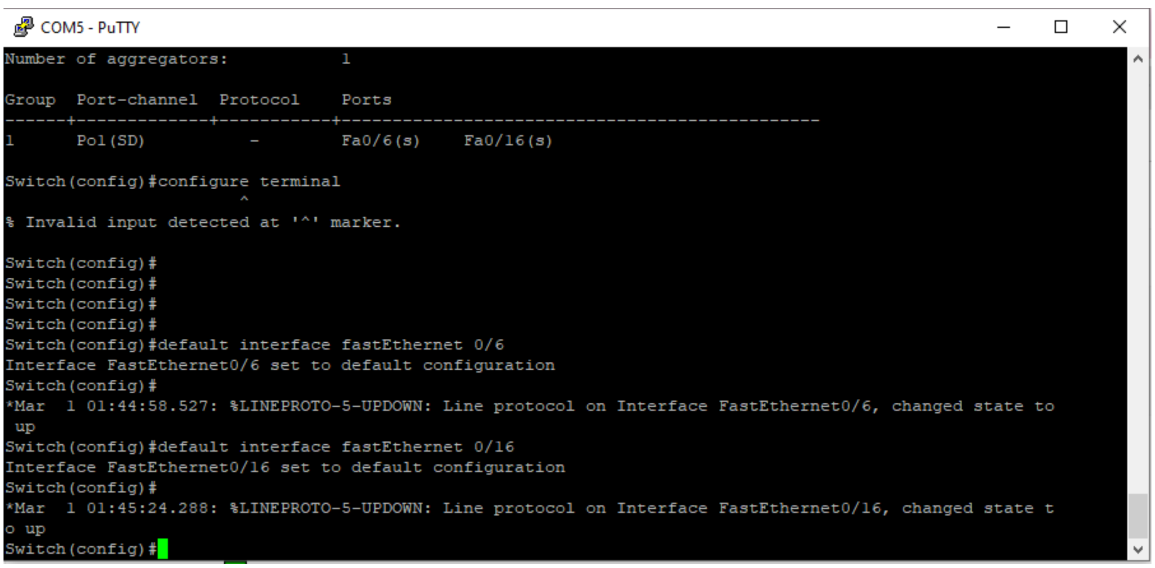
“En un Catalyst 2960 provoqué una incompatibilidad de configuración entre miembros de un EtherChannel, validé el estado con show vlan brief, show interfaces status y show etherchannel summary, y confirmé la suspensión automática de los puertos por mismatch de VLAN.”

En un NOC cuando se **cierra un incidente** hay dos opciones posibles:

| Opción | Qué significa | Cuándo se usa |
|---|---|---|
| **Restaurar servicio** | Volver la red al estado correcto | producción |
| **Documentar y dejar evidencia** | Mantener la falla para análisis | laboratorio / training |




- el EtherChannel queda **suspended**
- se intenta corregir
- pero el switch **no vuelve solo**
- entonces hay que **resetear interfaces**.



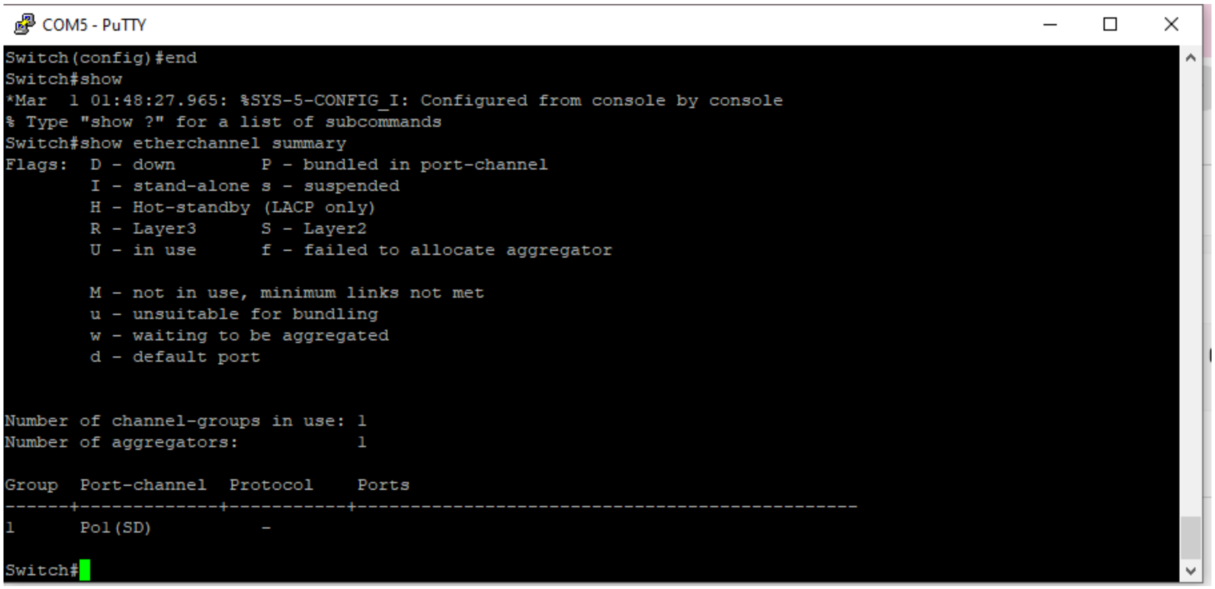

LINEPROTO-5-UPDOWN: Interface FastEthernet0/6 changed state to up
LINEPROTO-5-UPDOWN: Interface FastEthernet0/16 changed state to up

Eso significa:

 las interfaces fueron **reseteadas a configuración de fábrica**
 salieron del EtherChannel
 volvieron a estado normal

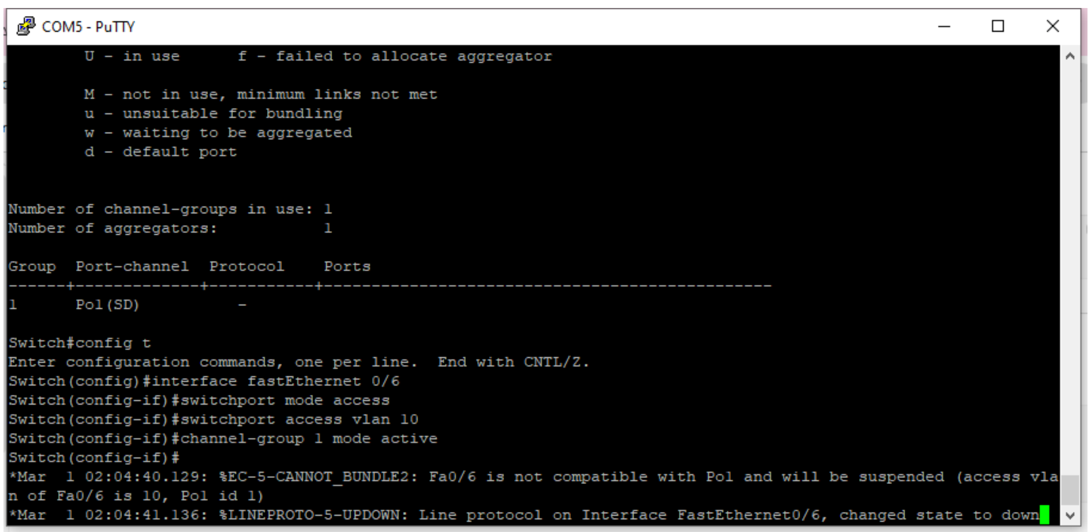
¿Qué haría un NOC cuando un **EtherChannel queda en estado inconsistente**?

las interfaces fueron **reseteadas a configuración**

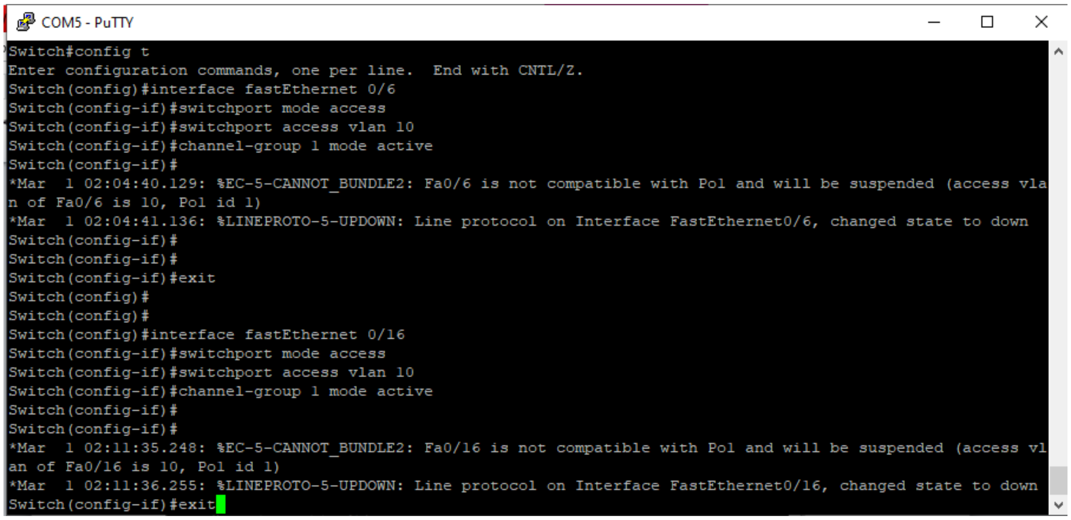



```
Switch(config)#end
Switch#show
*Mar  1 01:48:27.965: %SYS-5-CONFIG_I: Configured from console by console
% Type "show ?" for a list of subcommands
Switch#show etherchannel summary
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
1      Po1(SD)          -

Switch#
```

After resetting the interfaces to default configuration, Fa0/6 and Fa0/16 were removed from the EtherChannel, as the channel-group configuration was cleared.




```
        U - in use      f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------
1      Po1(SD)          -

Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fastEthernet 0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#channel-group 1 mode active
Switch(config-if)#
*Mar  1 02:04:40.129: %EC-5-CANNOT_BUNDLE2: Fa0/6 is not compatible with Po1 and will be suspended (access vla
n of Fa0/6 is 10, Po1 id 1)
*Mar  1 02:04:41.136: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to down
```



```
COM5 - PuTTY
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fastEthernet 0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#channel-group 1 mode active
Switch(config-if)#
*Mar  1 02:04:40.129: %EC-5-CANNOT_BUNDLE2: Fa0/6 is not compatible with Po1 and will be suspended (access vla
n of Fa0/6 is 10, Po1 id 1)
*Mar  1 02:04:41.136: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to down
Switch(config-if)#
Switch(config-if)#
Switch(config-if)#exit
Switch(config)#
Switch(config)#
Switch(config)#interface fastEthernet 0/16
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#channel-group 1 mode active
Switch(config-if)#
Switch(config-if)#
*Mar  1 02:11:35.248: %EC-5-CANNOT_BUNDLE2: Fa0/16 is not compatible with Po1 and will be suspended (access vl
an of Fa0/16 is 10, Po1 id 1)
*Mar  1 02:11:36.255: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/16, changed state to down
Switch(config-if)#exit
```

el EtherChannel **intentó formarse**
pero **los puertos quedaron suspendidos**.

## ¿Por qué pasa esto?

En tu log aparece otra vez:

EC-5-CANNOT_BUNDLE2
Fa0/6 is not compatible with Po1

Esto pasa porque **el Port-channel1 quedó con configuración vieja**.

Cuando hicimos:

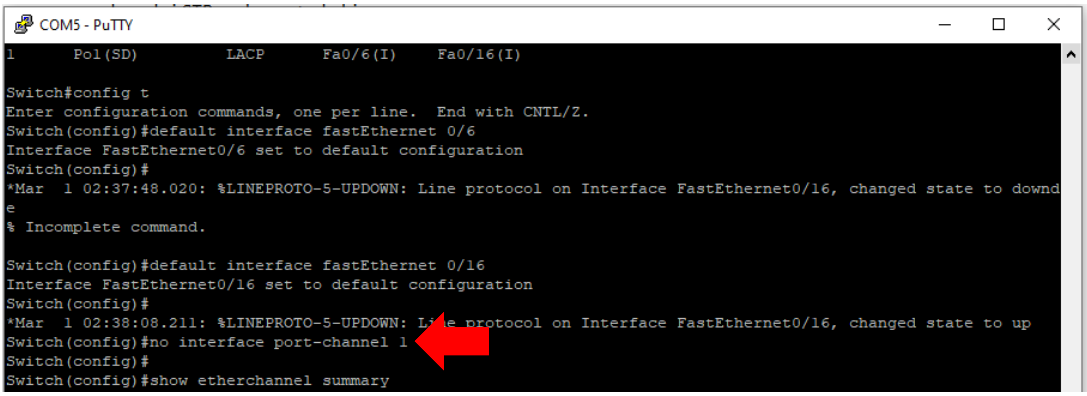
default interface fa0/6
default interface fa0/16

limpiamos los puertos, **pero no limpiamos el Po1**.

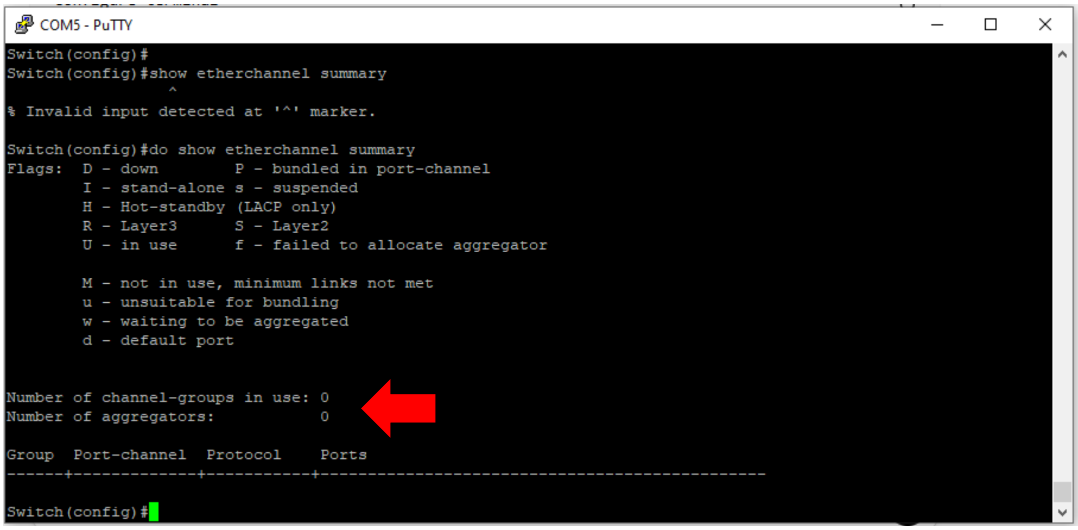
Entonces el switch cree que:

Po1 tiene una configuración distinta y suspende los puertos.

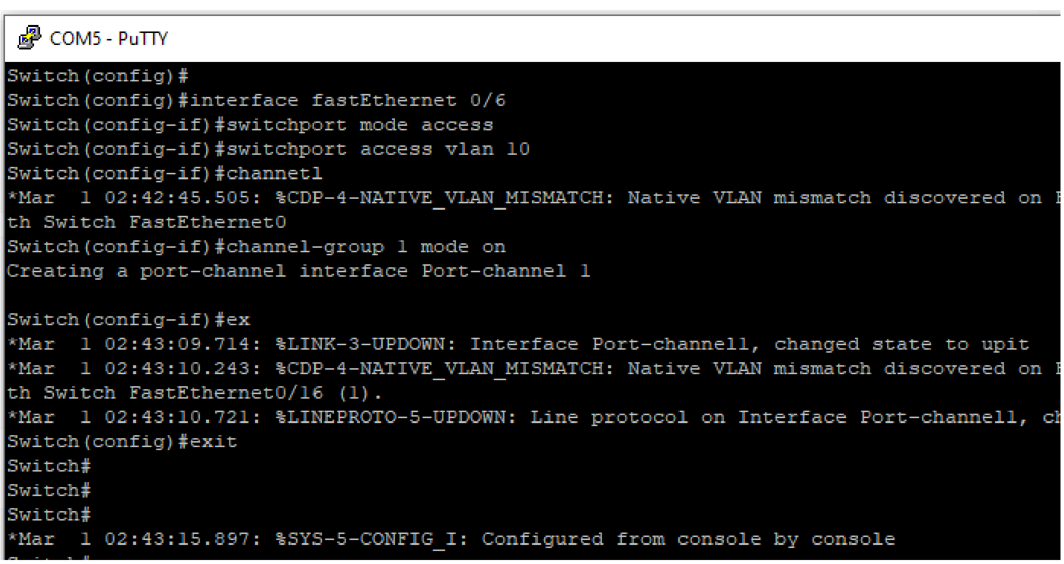
Configure terminal
default interface fastEthernet 0/6
default interface fastEthernet 0/16
no interface port-channel 1



```
COM5 - PuTTY
1      Po1(SD)          LACP      Fa0/6(I)     Fa0/16(I)

Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#default interface fastEthernet 0/6
Interface FastEthernet0/6 set to default configuration
Switch(config)#
*Mar  1 02:37:48.020: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/16, changed state to downd
e
% Incomplete command.

Switch(config)#default interface fastEthernet 0/16
Interface FastEthernet0/16 set to default configuration
Switch(config)#
*Mar  1 02:38:08.211: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/16, changed state to up
Switch(config)#no interface port-channel 1
Switch(config)#
Switch(config)#show etherchannel summary
```



```
COM5 - PuTTY
Switch(config)#
Switch(config)#show etherchannel summary
                 ^
% Invalid input detected at '^' marker.

Switch(config)#do show etherchannel summary
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 0
Number of aggregators:           0

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

Switch(config)#
```

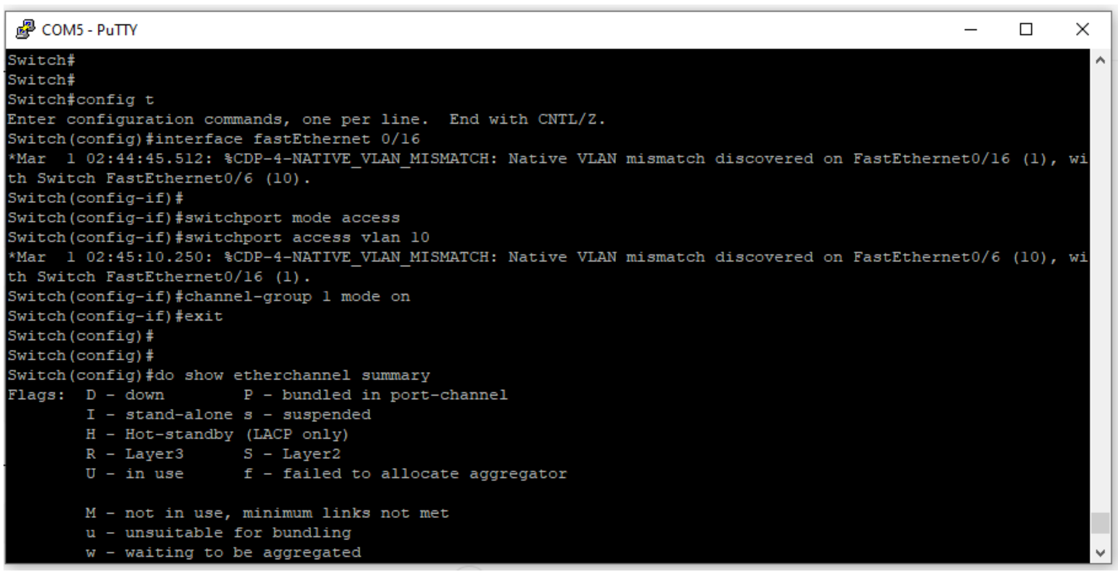


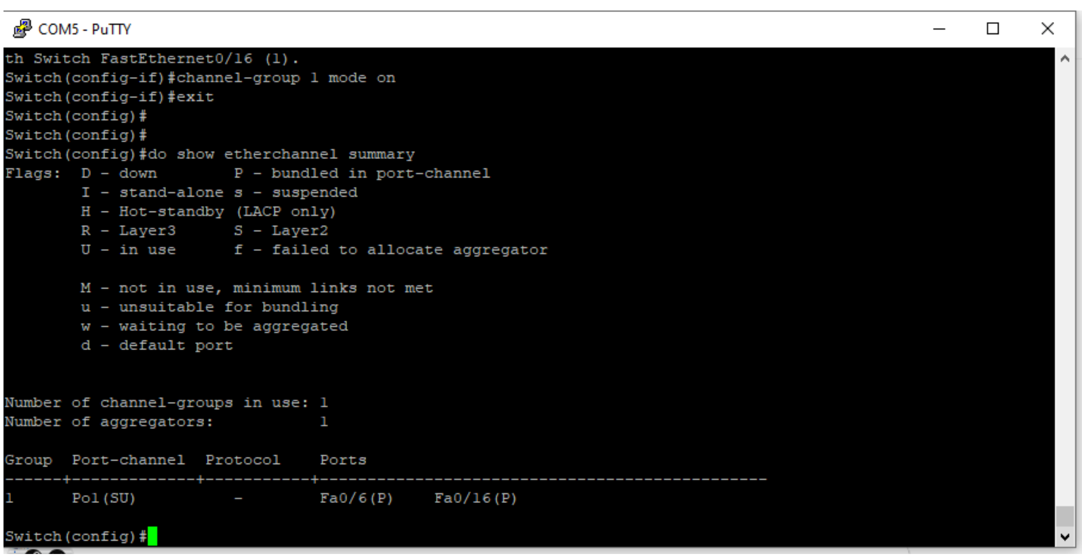
```
COM5 - PuTTY
Switch(config)#
Switch(config)#interface fastEthernet 0/6
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#channel1
*Mar  1 02:42:45.505: %CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on F
th Switch FastEthernet0
Switch(config-if)#channel-group 1 mode on
Creating a port-channel interface Port-channel 1

Switch(config-if)#ex
*Mar  1 02:43:09.714: %LINK-3-UPDOWN: Interface Port-channel1, changed state to upit
*Mar  1 02:43:10.243: %CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on F
th Switch FastEthernet0/16 (1).
*Mar  1 02:43:10.721: %LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, ch
Switch(config)#exit
Switch#
Switch#
Switch#
*Mar  1 02:43:15.897: %SYS-5-CONFIG_I: Configured from console by console
```

interface fastEthernet 0/6
switchport mode access
switchport access vlan 10

channel-group 1 mode on
exit

interface fastEthernet 0/16
switchport mode access
switchport access vlan 10
channel-group 1 mode on
exit




## Command "`show etherchannel summary`":

- `Po1(SU)`
- `Fa0/6(P)`
- `Fa0/16(P)`

## □ Interpretación técnica

- **S = Layer2**
- **U = in use → ACTIVO**
- **P = bundled → los puertos están dentro del canal**

**"I identified an EtherChannel failure caused by a VLAN mismatch between member interfaces. After resetting the interfaces and removing the old port-channel configuration, I reconfigured both ports with identical settings and successfully restored the bundle, verifying with `show etherchannel summary`."**

**Título:**
EtherChannel failure due to VLAN mismatch

### Tecnologías usadas

- VLAN
- EtherChannel
- LACP
- Forced EtherChannel
- Troubleshooting

### Comandos clave

show vlan brief
show interfaces status
show etherchannel summary
default interface
channel-group mode on

configure terminal
default interface fastEthernet 0/6
default interface fastEthernet 0/16
no interface port-channel 1

**RESOLUCION PASO A PASO / STEP-BY-STEP RESOLUTION**

| # | COMANDO / COMMAND | ACCION (ES) | ACTION (EN) |
|---|---|---|---|
| 1 | `default interface fastEthernet 0/6` | Resetea Fa0/6 a fabrica (limpia EtherChannel, VLAN, mode) | Resets Fa0/6 to factory (clears EtherChannel, VLAN, mode) |
| 2 | `default interface fastEthernet 0/16` | Resetea Fa0/16 a fabrica | Resets Fa0/16 to factory |
| 3 | `no interface port-channel 1` | Elimina Po1 — puertos quedan administratively down | Deletes Po1 — ports go administratively down |
| 4 | `interface fa0/6 → switchport mode access` | Configura modo acceso en Fa0/6 | Configures access mode on Fa0/6 |
| 5 | `switchport access vlan 10` | Asigna VLAN 10 a Fa0/6 (igual | Assigns VLAN 10 to Fa0/6 |

| | | | |
|---|---|---|---|
| | | que Fa0/16) | (matching Fa0/16) |
| 6 | `channel-group 1 mode on` | Re-agrega Fa0/6 al bundle forzado | Re-adds Fa0/6 to forced bundle |
| 7 | `interface fa0/16 → misma configuracion` | Replica configuracion identica en Fa0/16 | Replicates identical config on Fa0/16 |
| 8 | `show etherchannel summary` | Verifica bundle Po1 en estado U (in use) | Verifies bundle Po1 in state U (in use) |

## TROUBLESHOOTING DOCUMENTADO / DOCUMENTED TROUBLESHOOTING

| HALLAZGO / FINDING | ACCION CORRECTIVA / CORRECTIVE ACTION |
|---|---|
| Po1 DOWN y puertos en estado 'suspended' al iniciar el lab. / Po1 DOWN and ports in 'suspended' state at lab start. | Identificado como configuracion residual del lab anterior (EtherChannel). No afecto el lab actual pero requirio diagnostico antes de proceder. / Identified as residual config from previous lab. Did not affect current lab but required diagnosis before proceeding. |
| Error EC-5-CANNOT_BUNDLE2 persistio despues de hacer default interface en los puertos. / EC-5-CANNOT_BUNDLE2 persisted after running default interface on ports. | La causa fue que Po1 aun tenia configuracion vieja. Solucion: eliminar el Po1 con no interface port-channel 1 antes de reconfigurar los puertos miembro. / Cause: Po1 still had old config. Solution: delete Po1 with no interface port-channel 1 before reconfiguring member ports. |
| Tras no interface port-channel 1, los puertos quedaron administratively down. / After no interface port-channel 1, ports went administratively down. | Comportamiento esperado en Cisco IOS. Se reconfiguraron los puertos con mode access + VLAN 10 + channel-group 1 mode on. Puertos volvieron a estado up. / Expected Cisco IOS behavior. Ports reconfigured with mode access + VLAN 10 + channel-group 1 mode on. Ports returned to up state. |
| En laboratorio se documento la falla sin restaurar servicio en primera instancia (metodologia valida en training). / In lab, failure was documented without restoring service initially (valid training methodology). | Se documento el estado de falla (Incident Report) y luego se ejecuto la resolucion completa. Ambas opciones son validas en NOC: restaurar o documentar segun contexto. / Failure state documented (Incident Report), then full resolution executed. Both options valid in NOC: restore or document depending on context. |

## RESULTADO / RESULT

| | |
|---|---|
| **RESULTADO (ES)**<br>Falla de EtherChannel diagnosticada (VLAN mismatch), documentada en Incident Report formato NOC, y resuelta con exito. Po1 restaurado a estado activo. Syslog confirma interfaces up. Ticket cerrado. | **RESULT (EN)**<br>EtherChannel failure diagnosed (VLAN mismatch), documented in NOC-format Incident Report, and successfully resolved. Po1 restored to active state. Syslog confirms interfaces up. Ticket closed. |

**KEY TAKEAWAY**

*"En un EtherChannel todos los puertos miembro deben tener configuracion identica: mismo mode, misma VLAN, mismo tipo de negociacion. Un solo mismatch suspende todo el bundle. En NOC el proceso es: baseline → diagnostico → Incident Report → resolucion → verificacion → cierre de ticket. No se toca un puerto en produccion sin documentar primero el estado de falla."*

*"In an EtherChannel all member ports must have identical configuration: same mode, same VLAN, same negotiation type. A single mismatch suspends the entire bundle. In NOC the process is: baseline → diagnosis → Incident Report → resolution → verification → ticket closure. No production port is touched without first documenting the failure state."*

**NOC RELEVANCE — MINING / INDUSTRIAL ENVIRONMENT**

In a mining operation (e.g., Newmont Cerro Negro), EtherChannel links are used to aggregate bandwidth between core and distribution switches — carrying SCADA traffic, IP cameras, and VoIP simultaneously. A Po going down due to VLAN mismatch can silently disconnect multiple systems. A NOC engineer must: (1) quickly identify suspended ports via show interfaces status, (2) correlate with EC-5-CANNOT_BUNDLE2 in syslog, (3) determine if the mismatch is a config error or an unauthorized change, (4) open an Incident Report before touching anything in production, and (5) restore service following a tested procedure — exactly as documented in this lab.